06 · Case 4, part 1: Manaia's breathing

The words of this track, as it says them. Bold brown is the examiner.

Exam case 4. Panic attacks.

Manaia lives with an anxiety disorder, which sometimes shows up as panic attacks that affect her breathing. She describes a panic attack as a time when she breathes really, really hard and fast, and can't control her breathing at all. There's a link between breathing and the pH of the blood, because the respiratory system is one of the body's buffer systems. During a prolonged panic attack, Manaia might be at risk of respiratory alkalosis, and you'll need to remember why.

Case 4 is the other big one, so it's three tracks as well. First, the mechanics of breathing.

Four marks. Bullet point the steps involved in Manaia's inhalation.

The hook: contract, bigger, lower pressure, air in.

The diaphragm contracts, so its dome lowers. The external intercostal muscles contract, moving the ribs up and out. That enlarges the thoracic cavity, and reduces the pressure in the intrapleural cavity. That's Boyle's law. The lungs follow, so the intrapulmonary volume increases, and the intrapulmonary pressure falls. Air flows in from the atmosphere, from high pressure to low.

Four marks. Bullet point the steps involved in Manaia's exhalation.

The hook: relax, recoil, higher pressure, air out.

The diaphragm relaxes, so its dome rises. The external intercostals relax, so the ribs move down and in. The elastic lungs recoil passively, reducing the intrapulmonary volume. Intrapulmonary pressure rises: Boyle's law again. And air flows out of the lungs, from the higher intrapulmonary pressure to the lower atmospheric pressure.

Two marks. Manaia feels tired after a panic attack, because forced breathing takes more muscular effort. Which muscles are used in forced breathing? Be specific.

Breathing in. At rest, the diaphragm and the external intercostals contract. When it's forced, the neck muscles join in: the sternocleidomastoid, and the scalenes.

Breathing out. At rest it's passive: the diaphragm and external intercostals just relax. When it's forced, the abdominal muscles and the internal intercostals contract.

So the hook: breathing in, the neck lifts. Breathing out, the abs and the internal intercostals squeeze.

Three marks. What are some of the factors that influence pulmonary ventilation, making it easier or harder, and how do they do this?

The hook: blocked airways, stiff lungs, and a lost pleural seal.

Blocked airways. Obstruction by fluid, mucus or swelling, as in COPD and asthma, narrows the airways. That raises the resistance to airflow, and the muscular effort needed to breathe in and out.

Stiff lungs. Lung compliance is the ease with which the lungs expand, and scar tissue reduces it. So do problems with the thoracic cage, like an inflexible rib cage or a diaphragm problem. And so do problems in the alveoli: a lack of surfactant, so the alveoli don't expand, or fluid building up in them.

The pleural seal. The negative pressure in the pleural space couples the lungs to the chest wall. Air in the pleural cavity, a pneumothorax, breaks that seal, and the lung collapses. That's atelectasis.

Once more. Manaia's breathing.

Inhalation: the diaphragm and external intercostals contract, the thoracic cavity enlarges, the intrapleural and then intrapulmonary pressures fall, by Boyle's law, and air flows in from high pressure to low. Exhalation: they relax, the elastic lungs recoil passively, intrapulmonary pressure rises, and air flows out. Forced breathing in adds the sternocleidomastoid and scalenes; forced breathing out uses the abdominal muscles and internal intercostals. Ventilation gets harder with blocked airways, with stiff lungs from scar tissue or no surfactant, and with a lost pleural seal: a pneumothorax collapses the lung.